

Heat Transfer Modeling of Thermal Spraying Processes: An Insight into Gas Turbine Engine Safety and Performance



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Abstract

Gas turbine engines are reliable pieces of machinery that are used to benefit from a work output either in industrial settings to power homes and cities, or for propelling aircraft. Operation of those engines in particle dense environments such as in environments where there is a lot of dust, sand or volcanic ash can lead to rapid deterioration in engine performance and catastrophic outcomes. One of the main phenomena that attracts particular interest is the phase change and impingement process of molten sand particles on the turbine guide vanes and Thermal Barrier Coatings (TBCs). Extensive work has been done to model the effects of thermal spraying analytically and experimentally on substrates of different thermal conditions and geometries. However, most of the work models particles other than sand or glassy (CMAS) particles in general, and models the impingement process on a flat substrate, while fewer works track CMAS particle thermal spraying on a curved substrate. This work will investigate the current state-of-the-art work achieved in terms of understanding the physical properties and behavior of thermally sprayed particles as it pertains to gas turbine settings. The work will highlight analytical and numerical models aimed to study the heat transfer phenomena of the particle in question by modeling the phase change through solidification or otherwise and the impingement processes. This work will also cover future work and improvements projected in this field and will tackle the shortcomings of the current models and data available.

Introduction

Background

Gas turbines are expensive pieces of machinery designed to accelerate and expand air to generate flow work. This work output could be utilized in powering cities and homes in the action of industrial gas turbines situated in industrial power plants or it could be utilized in propelling modern aircraft.

Effective and reliable operation of gas turbines and especially gas turbine jet engines impose major implications on public and environmental safety and economic stability. For example, in 2010, over 100,000 flights were terminated by the European aviation authorities after the Eyjafjallajökull volcanic eruption ensued in Iceland leading to economic losses up to \$ 1.7 billion in that period [1].

To maximize fuel economy and maintain environmental stability, gas turbine engines are designed to operate at very high temperatures, maximizing the thermal efficiency of the engine. Turbine blades, for instance, can operate safely when the Combustor Exit Temperature (CET) reaches around 1700 ° C due to the presence of a Thermal Barrier Coating (TBC) which is a protective coating made from ceramic powders that is deposited onto the turbine blade through various thermal spraying techniques such as Plasma Spraying (PS) or Electron Beam-Physical Vapor Deposition (EB-PVD) [2][3]. These TBCs provide adequate thermal insulation to the blade's surface maintaining the thermal and structural nature of the blade.

Gas turbines operating in particle-dense environments such as sandy environments like deserts or in fly zones near a volcanic eruption can lead to significant damage to internal components by the action of erosion or deposition. The latter being of more significance and interest due to the rapid damaging effects taking place in a very short period (Figure 1) [4]. Such events can lead to mid-flight engine failures due to stalling similar to the accident occurring with a British Airways flight in 1982 when a Boeing 747 had all its four engines stall due to the ingestion of volcanic ash particles.

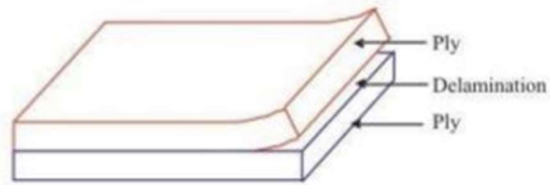


Figure 1: Sand ingestion and deposition on turbine guide vanes

The solidification of ingested airborne particles such as dust and volcanic ash mainly categorized as Calcium–Magnesium–Alumino–Silicate (CMAS) particles happens when particles typically smaller than 75 microns bypass the installed particle separators and make their way through the combustion chamber. Depending on the CET pertaining to the combustion chamber in question, the ingested CMAS particles can either soften or fully melt. Particularly, molten droplets are of more interest as the dynamics of semi-molten particles have been covered extensively in the literature.

The molten droplets can cause thermo-mechanical damages such as infiltration, thermal expansion/contraction (thermal cycling) and other failures mechanisms such as delamination or spallation shown in Figure 2 [5][6].

DELAMINATION



Separation of adjacent layers due to weakening of interface layer between them.

EXPERIMENTAL SHOCK-PRODUCED SPALLATION

QUARTZ SPHERES $\varnothing$ 14 mm IN EPOXY MATRIX

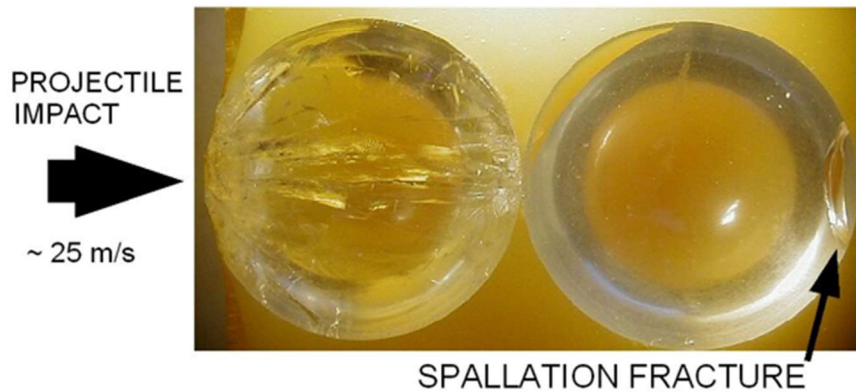


Figure 2: Delamination (up) and spallation (bottom) thermo-mechanical effects on TBCs

Motivation

This work aims to characterize, model, and understand the heat transfer processes and thermal variations occurring when a molten droplet impinges and solidifies onto a substrate. The work will present solutions mainly concerned with the conduction aspect of the heat transfer process (low Peclet (Pe) number), namely the mere contractual interaction between the droplet and the substate and will disregard the heat transfer effects on the particle itself from convection and radiation.

This understanding helps engineers predict the thermal and structural behavior of the substate by tracking effects such as the infiltration rate, change in droplet viscosity, and the propagation of the structural deformations described earlier, which aids in designing more resilient and capable TBCs for future gas turbine iterations.

Methodology

E-MPS Numerical Simulation and Experiment

The Explicit-Moving Particle Simulation (E-MPS) is a Lagrangian based computational method used to model and simulate complex flow fields [7]. Fukudome et al utilize this modeling technique to simulate the flow and solidification process of a tin (Sn) droplet on a non-isothermal, stainless steel substrate [8].

To construct the model, the Sn droplets were assumed to be incompressible, the convective and radiative heat transfer effects between the ambient air and the imprinting Sn droplet were neglected, any air voids or gaps forming during contact between the Sn droplet and the substrate were neglected, the side and bottom surfaces of the substrate (the surfaces that are not in direct contact with the droplet) were assumed to be in adiabatic condition, and finally the volumetric change resulting from the phase transition was also neglected [8].

The continuity, Navier-Stokes, and momentum conservation equations are represented in Eq (1), Eq (2), and Eq (3) respectively below, are solved in the model to obtain relevant flow and thermal parameters such as the density (ρ), time (t), pressure(p). kinematic viscosity (ν), velocity vector ($\mathbf{u}$), external force consisting of the gravitational component and the component from surface tension ($\mathbf{G}$), the temperature (T), specific heat capacity (C_p), thermal conductivity (k), and heat transfer rate (q) [8].

$$\frac{D\rho}{Dt} = 0 \quad (1)$$

$$\frac{D\mathbf{u}}{Dt} = \frac{-1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{G} \quad (2)$$

$$\rho C_p \frac{DT}{Dt} = k \nabla^2 T + q \quad (3)$$

To successfully model the heat transfer effects on the stainless steel substrate when it is assumed to be non-isothermal, a coupling method was introduced to the model as a boundary treatment by projecting dummy droplets onto the substrate surface to improve the numerical accuracy of the simulation when a non-isothermal condition is imposed. The coupling method accuracy was further validated by projecting the results of the simulation into a graph showing the temporal temperature distribution as a function of the radial width of the splat against the analytically obtained temperature distribution given in Eq (4) [8]. The graph is displayed in Figure 3.

$$T(x) = T_0 - (T_s - T_0) \left\{ 1 - \operatorname{erf} \left(\frac{x}{2\sqrt{\frac{kt}{\rho c_p}}} \right) \right\} \quad (4)$$

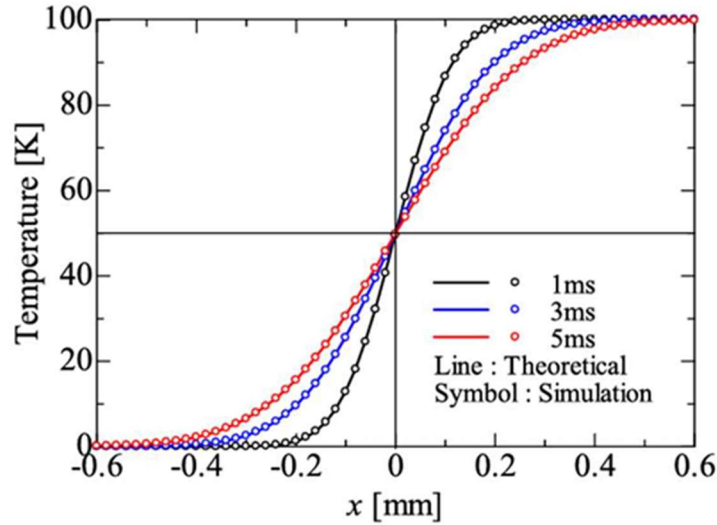


Figure 3: Temporal temperature distribution of substrate as a function of the radial width of the splat presenting simulation and analytical results

The experimental setup of the problem with the substrate and droplet conditions are displayed in Figure 4 (top). The setup was exactly duplicated in the model to ensure the best and most accurate results possible are obtained from the comparative analysis with the experiment. The numerical and experimental evolution of the imprisonment and solidification processes is shown in Figure 4 (bottom) in a simultaneous fashion [8].

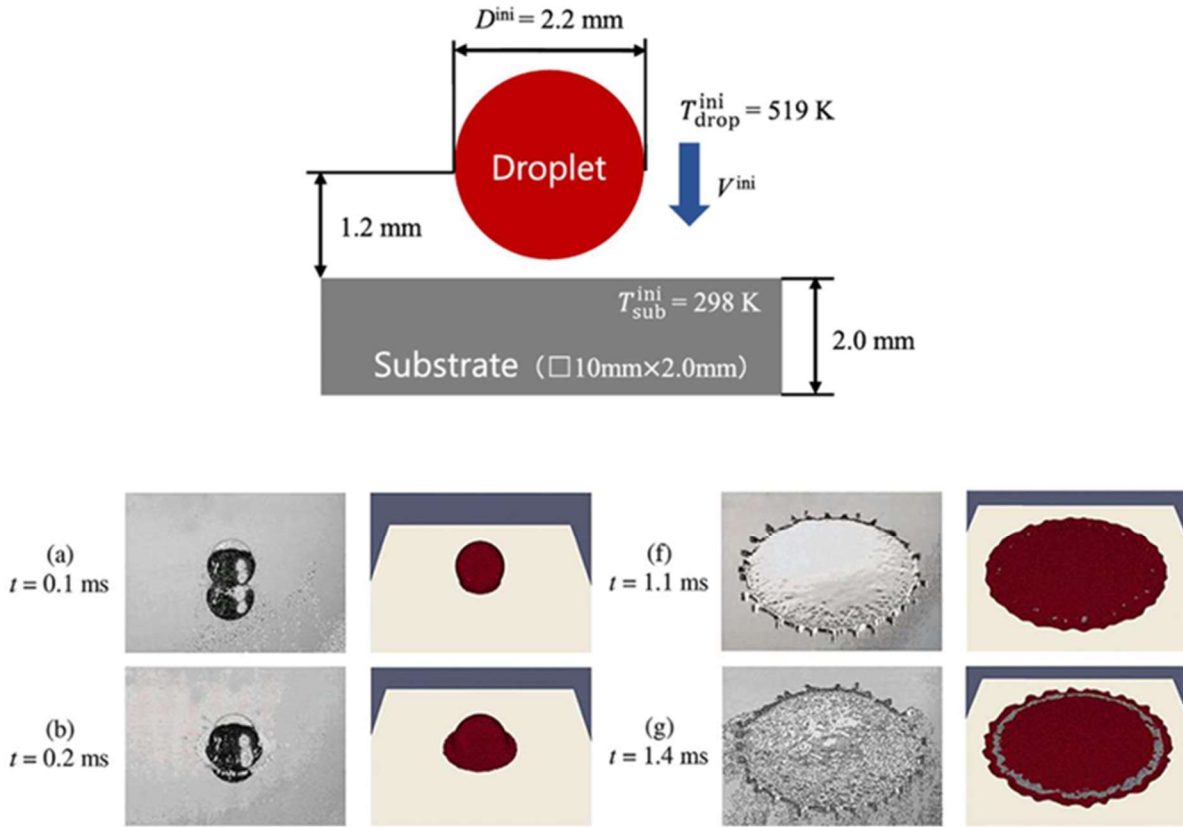


Figure 4: Problem setup (top), simultaneous impingement evolution for model and experiment

Other Solution Approaches

Modeling the thermal-fluidic behavior of particle impingement is usually performed using a main model that establishes the flow field and a coupling method introduced to account for the heat transfer effects. Usually, the heat transfer is directly related to the flow for if the flow field is modeled precisely, the heat transfer results would be correctly obtained. Many models exist that either use a Eulerian approach or a Lagrangian approach such as Volume of Fluids (VOF) model, Large Eddy Simulations (LES) model, and the (E-MPS) mode to track the velocity profile of the particle. However, models like the Coupled Eulerian-Lagrangian (CEL) method combine the strengths of the two approaches [1]. Other models use Finite Element Method (FEM) with coupling techniques to capture the thermal-fluidic aspects [3]. The correct use of any of these models or the establishment of a new model considers the level of complexity needed, and the specific problem tackled. For example, models used to study semi-molten or softened particles like the Smoothed Particle Hydrodynamics (SPH) method would not be adequate in a lot of areas to model fully molten droplets [1].

Results and Discussion

E-MPS Numerical Simulation Method

The captured temperature distribution profile of the splat as it impinges and solidifies on the substrate is outlined in Figure 5. As seen in Figure 5, the hottest region on the splat is situated around the center but not at the center directly. Fukudome et al argue that this is caused due to the higher convection rate in the ring-shaped region from the presence of more liquid particles while the convection being weak on the other at the center where most of the particles already solidified [8].

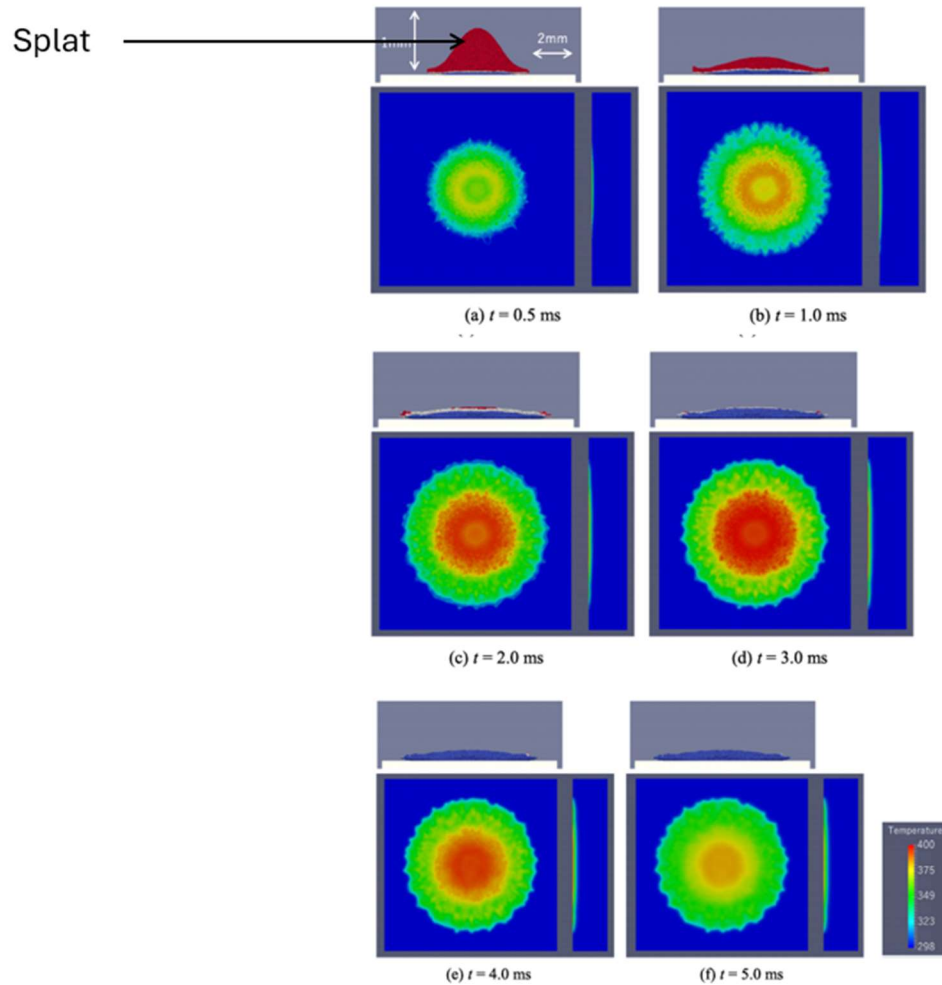


Figure 5: Time evolution of splat impingement and solidification with temperature distribution of substrate

The key finding from this observed result lies in the fact that the hotter regions (around the center) and have a thin film thickness and the regions where solidification is more pertinent (at the center and at the rim) are thicker in film thickness. These results indicate that thicker films enhance the process of solidification for the interaction between a fully molten droplet on a flat, non-isothermal substrate. The discussed findings have implications towards an improved understanding the complexity of the droplet and solidification process from its heat transfer aspects.

Other Noteworthy Results – Viscosity Effect

Figure 6 shows the relationship between the temperature distribution and the dynamic viscosity of 4 different variants of softened volcanic ash particles. As seen in Figure 6, the viscosity is inversely proportional to temperature. From those findings, it can be observed that at lower viscosities and higher particle temperatures, the dissipation rate will be higher leading to more rapid solidification. Additionally, during the initial stages of the impingement where the pressure gradient of the impingement is pertinent, lower viscosities improve the infiltration rate on the TBCs [1] [9].

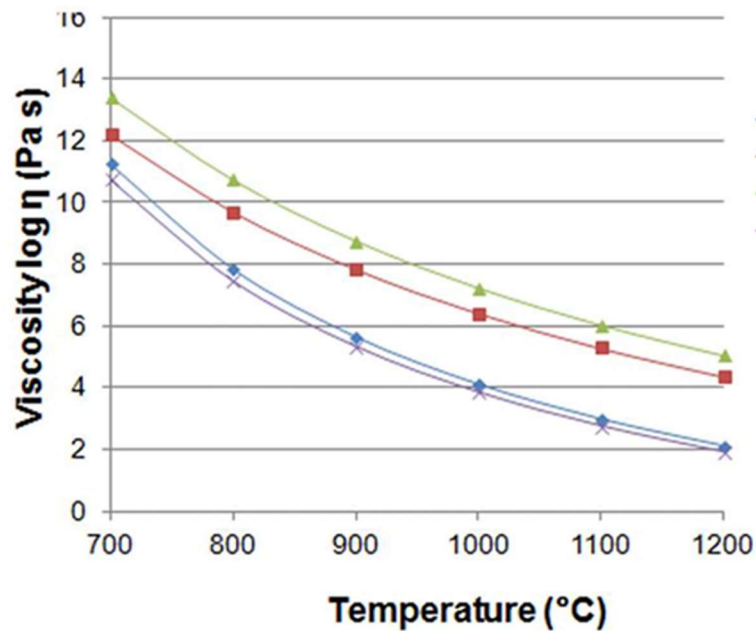


Figure 6: Dynamic viscosity vs temperature distribution for 4 volcanic ash softened particle variants

Conclusion and Next Steps

The following work attempted to outline the thermal effects of particle deposition and solidification on the turbine blades and TBCs by presenting a model that describes the solidification of a Sn particle on a flat, stainless steel non-isothermal substrate. The model portrayed that the solidification process happens more towards the center than around it where the film thickness is thin. However, the model poses limitation because it does not study the isothermal case and it is tested on temperature much lower than the expected onto the CET. The analysis nevertheless gave insight into a better understanding of the physics and heat transfer involved in such problems, which in turn aids engineers to design more capable TBCs.

Next steps include the development of more models that are more capable as running actual experiments is a costly method, Additionally, investigating other kinds of thermal barrier coatings and the various ways they can be thermally sprayed and their microstructure and how it is affected is crucial to develop more robust technique for resisting the damage endured from the deposition effects.

Nomenclature

TBC: Thermal Barrier Coating

CMAS: Calcium–Magnesium–Alumino–Silicate

CET: Combustor Exit Temperature

PS: Plasma Spraying

EB-PVD Electron Beam-Physical Vapor Deposition

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